

Docker

1. Architectural Core & Value Proposition

Definition & Overview

From what I learned, Docker is a containerization platform that packages an application together with its runtime, libraries, configuration files, and dependencies. I understood that Docker solves the common 'works on my machine' problem because the same container can run consistently in development, testing, and production. Docker containers are lightweight compared to virtual machines because they share the host operating system kernel. I also explored concepts such as Docker images, containers, Dockerfiles, Docker Hub, networking, and volumes. These concepts helped me understand how modern applications are deployed efficiently. I feel this concept is highly relevant in today's software industry because organizations demand faster delivery, reliability, security, and scalability. Through this module I gained a practical understanding of why these technologies are widely adopted and how they support enterprise application development.

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Why? (Problem Solved)

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How it Solves the Problem

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Key Benefits

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2. Foundational Pillars & Key Concepts

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3. Core Concepts

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4. Engineering Tooling & Implementation

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5. Operational Best Practices, Risks & Security

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6. Key Takeaways

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